

Research Plan

Advanced Threat Monitoring for RobotLab

Fontys University of Applied Sciences

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Authors: Team 5

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1 Introduction

This document presents the Research Plan for the project Advanced Threat Monitoring for RobotLab. The goal of this document is to clearly explain what the project is actually about, what problem we are trying to solve, which research questions we want to answer, and how we plan to approach the research.

Our project, is focused on researching how threat monitoring can be improved in this environment. The main idea is that we investigate how a advanced monitoring approach can be designed, with the goal of creating a reference architecture for threat detection in RobotLab. This is a Blue Team focused project, which means that we focused on detecting, monitoring, responding and analysing suspicious activity.

This is mainly a research project, so the main focus is on analysing the problem, exploring possible solutions, and designing a suitable approach. Based on the results of the research, the project will also include an implementation part. In this way, the research phase will guide the implementation and help make better justified technical decisions.

2 Problem Statement

RobotLab is currently relying on AWS GuardDuty for threat detection. While this already provides a basic level of visibility, it is focused on high level logs and does not give a deeper monitoring that it is needed for a more advanced and automated environment. For example, we don't have a Deep Packet Inspection, which means that some attacks may stay hidden inside network traffic. AWS GuardDuty also does not provide enough visibility into runtime behaviour in Kubernetes environments, making detection of advanced threats harder.

For RobotLab this is a big issue because it is running in a cloud based environment where different systems, services, and processes are connected. In this environment, attackers may target infrastructure and also sensitive research data. If threat detection stays high level, malicious activity might be missed.

Another challenge is that cloud security services become expensive as the environment grows, giving a limited customization and flexibility. Because of this, we clearly need to research how RobotLab can move towards a more comprehensive and self hosted monitoring architecture that provides deeper visibility, better detection, and more control over security data.

3 Research Goal

The goal of this project is to research on how we can build a advanced threat monitoring environment in RobotLab's and to design a suitable approach for it. The project aims to investigate which monitoring methods, tools, and techniques that provide stronger detection capabilities and better visibility. Based on research results, the project will work towards a architecture for advanced threat monitoring in RobotLab, followed by implementing the most suitable approach.

4 Expected Products

The expected products of this project are:

1. A comparison of relevant monitoring tools, methods, and techniques
2. A reference architecture for advanced threat monitoring in RobotLab
3. An implementation of the most suitable solution, or of selected components of that solution
4. Documentation of the research process, technical choices, and obtained results
5. Final conclusions and recommendations for future improvements

5 Research Questions

5.1 Main Research Question

How can an advanced hybrid monitoring system, integrating HIDS and NIDS, and leveraging SDN for dynamic control, be designed and implemented to protect RobotLab's AWS and on premise infrastructure from advanced cyberattacks?

5.2 Sub Questions

1. How can the monitoring system combine kernel level visibility through eBPF, run-time monitoring, and deep packet inspection to achieve effective threat detection across the infrastructure?
2. How can the monitoring system automatically respond to detected threats in order to contain attacks and minimize damage?
3. How should the defense layers be structured and distributed throughout the system to ensure both efficiency and security?

4. How can the attack surface be examined and how can the solution be validated?
5. How can data from HIDS and NIDS be correlated to accurately detect cyberattacks within the infrastructure?

6 Scope

6.1 In Scope

The following aspects are within the scope of this project:

- Research into advanced threat monitoring for RobotLab
- Comparison of relevant monitoring tools, methods, and techniques
- Design of a reference architecture for advanced threat monitoring
- Implementation of the most suitable solution, or selected components of that solution
- Documentation of research findings, technical choices, and results

6.2 Out of Scope

The following aspects are outside the scope of this project:

- The future on premises environment planned for RobotLab
- Infrastructure design, development, and management
- Broader system architecture changes unrelated to threat monitoring

7 Research Approach

This project follows the DOT (Development Oriented Triangulation) Framework, which is the standard research methodology used at Fontys University of Applied Sciences for ICT projects. The DOT Framework provides a structured way to combine different research strategies and methods in order to answer our research questions with sufficient rigour and validity.

The core idea behind the DOT Framework is **triangulation**: each research question should be answered using multiple methods from different research strategies, so that findings can be cross-validated and the conclusions are stronger. The framework defines five research strategies, each of which is described below along with how it applies to this project.

7.1 Research Strategies

7.1.1 *Library*

Library research involves studying existing knowledge through literature, documentation, and publicly available resources. In this project, Library research is used to build a theoretical foundation on advanced threat monitoring concepts, including HIDS, NIDS, eBPF-based observability, deep packet inspection, SDN-based response mechanisms, and data correlation techniques. Specific methods used include:

- **Literature Study** – Reviewing academic papers, industry whitepapers, and technical documentation on threat monitoring architectures, eBPF security tooling (Cilium, Falco), NIDS platforms (Suricata, Zeek), SIEM correlation, and SDN security frameworks.
- **Available Product Analysis** – Comparing existing monitoring tools and platforms to understand their capabilities, limitations, and suitability for the RobotLab environment.
- **Best, Good & Bad Practices** – Identifying established design patterns and anti-patterns for cloud monitoring infrastructures.

7.1.2 *Field*

Field research involves gathering information from the real-world context in which the project takes place. For this project, the field is RobotLab’s existing AWS infrastructure and the stakeholders involved in its security operations. Specific methods used include:

- **Problem Analysis** – Analysing the current state of RobotLab’s threat detection capabilities, identifying gaps in coverage, and understanding the specific risks of the environment.
- **Infrastructure Analysis** – Mapping the current RobotLab infrastructure, including network topology, service architecture, existing security tooling, and data flows, to inform the design of the monitoring architecture.

7.1.3 *Lab*

Lab research involves testing ideas and validating solutions in a controlled environment. This strategy is central to the implementation and validation phases of the project. Specific methods used include:

- **Component Testing** – Testing individual monitoring components (eBPF probes, Suricata rulesets, log collectors, correlation rules) in isolation to verify their functionality and performance.
- **System Testing** – Testing the integrated monitoring architecture as a whole to ensure that HIDS and NIDS data flows, correlation logic, and automated response mechanisms work together correctly.

7.1.4 *Showroom*

Showroom research involves presenting intermediate and final results to stakeholders and experts for feedback and validation. Specific methods used include:

- **Peer Review** – Presenting research findings, architecture designs, and implementation progress to fellow students and team members for critical feedback.
- **Expert Review** – Consulting with the course instructors and RobotLab stakeholders, to validate the architecture design and technical decisions.
- **Prototype Demonstration** – Demonstrating working prototypes of the monitoring system to stakeholders in order to gather feedback on functionality and coverage.

7.1.5 *Workshop*

Workshop research involves exploring and creating solutions through hands-on activities. This strategy supports the design and prototyping phases. Specific methods used include:

- **Prototyping** – Building iterative prototypes of individual monitoring components and the overall architecture to explore technical feasibility and refine the design.
- **Design Pattern Research** – Investigating established architectural patterns for security monitoring systems (defense-in-depth layering, SIEM pipeline patterns, SDN control loops) and applying them to RobotLab’s context.
- **Gap Analysis** – Comparing the current monitoring capabilities of RobotLab with the desired state defined through the research, in order to identify what needs to be built.

7.2 DOT Framework Mapping per Sub Question

The table below shows which DOT Framework strategies and methods are applied to each sub question, ensuring triangulation across the research.

SQ	Strategy	Method	Purpose
1	Library	Literature Study	Study eBPF, DPI, and run-time monitoring theory
1	Library	Available Product Analysis	Compare tools: Cilium, Tetragon, Falco, Suricata, Zeek
1	Lab	Component Testing	Test individual monitoring probes
1	Workshop	Prototyping	Build proof-of-concept for kernel-level and DPI pipelines
2	Library	Literature Study	Research automated response and SDN-based containment
2	Library	Best, Good & Bad Practices	Learn from existing SOAR and SDN response patterns
2	Lab	System Testing	Test automated response workflows end-to-end
2	Workshop	Design Pattern Research	Explore control loop and response orchestration patterns
3	Library	Literature Study	Study defense-in-depth and layered security models
3	Field	Infrastructure Analysis	Map current RobotLab infrastructure and data flows
3	Workshop	Gap Analysis	Compare current state to desired defense layering
3	Showroom	Expert Review	Validate layering decisions with stakeholders
4	Field	Problem Analysis	Identify attack surface of the RobotLab environment
4	Lab	Security Testing	Execute controlled attack simulations
4	Showroom	Peer Review	Present validation results for critical review

SQ	Strategy	Method	Purpose
4	Showroom	Expert Review	Validate findings with security experts
5	Library	Literature Study	Research SIEM correlation and data fusion techniques
5	Library	Available Product Analysis	Evaluate correlation engines and SIEM platforms
5	Lab	Component Testing	Test correlation rules against sample data
5	Lab	System Testing	Validate correlation accuracy in integrated system

8 Methodology

This project will be carried out using an Agile methodology. This means the work will be done in smaller iterations, where progress is reviewed regularly and adjustments can be made when needed. This approach is useful for a project like this because both the research and the implementation may develop over time, depending on new findings, technical challenges, and feedback during the process.

Within this Agile way of working, the project is still divided into several methodology phases. It starts with a literature review and an analysis of the RobotLab environment in order to understand the current situation, the main challenges, and the requirements for advanced threat monitoring. Based on these findings, a system design will be created in the form of a reference architecture.

After the design phase, the project will continue with implementation, where the most suitable solution, will be applied. The implemented solution will then be tested to check whether it works as intended, validated to determine whether it actually addresses the problem in a meaningful way, and finally evaluated based on the project goal and research questions.

Because the project follows an Agile methodology, these phases will not always be completed in one strict linear order. Instead, some activities may overlap or be repeated in multiple iterations. In this way, the project remains structured, while also allowing flexibility to improve the solution step by step during the process.

8.1 Literature Review

The project starts with a literature review in which relevant academic sources, technical documentation, and best practices are studied. This provides the theoretical foundation for the project and helps identify relevant tools, methods, and techniques for advanced threat monitoring.

8.2 Infrastructure Analysis

After that, the RobotLab environment will be analysed from a security monitoring perspective. The goal of this phase is to understand the environment, identify important monitoring needs, and determine where the main limitations or blind spots are. This analysis is only focused on monitoring and detection needs, not on infrastructure development itself.

8.3 System Design

Based on the literature review and the infrastructure analysis, a system design will be created. This design will take the form of a reference architecture for advanced threat monitoring in RobotLab and will describe how the selected monitoring components can work together.

8.4 Implementation

After the design phase, the project will move to implementation. In this phase, the most suitable solution, or selected components of that solution, will be applied in practice. This implementation will be guided by the outcomes of the earlier research phases.

8.5 Testing

Once the implementation is in place, testing will be carried out to check whether the solution works correctly. This includes verifying whether the monitoring components function as expected and whether they provide useful results.

8.6 Validation

The validation phase focuses on determining whether the implemented solution actually contributes to better advanced threat monitoring in RobotLab. This means checking whether the solution is not only technically correct, but also relevant and effective in the project context.

8.7 Evaluation

In the final phase, the project results will be evaluated. This includes reflecting on the research process, the implementation, and the final outcome, as well as determining to what extent the research questions and project goal have been addressed.

9 Planning

The planning of this project will be based on an Agile approach and will therefore be divided into sprints. By working in sprints, the team can structure the project into smaller and more manageable parts, while also creating regular moments to review progress and make adjustments when needed.

Each sprint will focus on a specific part of the project. In the earlier sprints, the main focus will be on research activities such as the literature review, analysing the RobotLab environment, and identifying suitable monitoring solutions. In the middle part of the project, the sprints will focus more on the design of the reference architecture and on preparing the implementation. In the later sprints, the focus will shift towards implementation, testing, validation, and evaluation.

At the end of each sprint, the results will be reviewed to determine what has been achieved, what still needs to be improved, and what the priorities are for the next sprint. This makes it possible to work in a structured way while still keeping enough flexibility to respond to new findings or technical challenges during the project.

9.1 Project Phases

Sprint	Activity	Expected Output
1	Project orientation, literature review, and initial exploration of advanced threat monitoring concepts	Initial understanding of the project context, theoretical foundation, and first overview of relevant concepts and technologies
2	Infrastructure analysis and identification of monitoring requirements for RobotLab	Clear overview of the project environment, identified monitoring needs, and relevant security requirements
3	Comparison of monitoring tools, methods, and techniques, and initial design decisions	Structured comparison of possible solutions and first design choices for the monitoring approach

4	Development of the reference architecture for advanced threat monitoring	Reference architecture describing the proposed monitoring approach for RobotLab
5	Implementation of the most suitable solution, or selected components of that solution	First practical implementation of the proposed monitoring solution or selected components
6	Testing, validation, evaluation, and final documentation	Tested and validated solution, evaluation of results, and final project documentation

9.2 Timeline

Week	Activity
Week 1	Project orientation and planning
Week 2	Literature review
Week 3	Literature review and exploration of advanced threat monitoring concepts
Week 4	Infrastructure analysis
Week 5	Identification of monitoring requirements
Week 6	Comparison of monitoring tools, methods, and techniques
Week 7	Comparison of solutions and initial design decisions
Week 8	Development of the reference architecture
Week 9	Development of the reference architecture
Week 10	Preparation for implementation
Week 11	Implementation of the selected solution or components
Week 12	Continued implementation
Week 13	Testing of the implemented solution
Week 14	Validation of the implemented solution
Week 15	Evaluation of results
Week 16	Final documentation
Week 17	Final review and completion of the project

10 Risk Analysis

Risk	Impact	Probability	Mitigation
eBPF tooling incompatibility with RobotLab's kernel or AWS environment	High	Medium	Verify kernel version compatibility early in Sprint 1 and identify fallback tools
AWS access limitations restricting DPI or runtime monitoring capabilities	High	Medium	Clarify access boundaries with RobotLab stakeholders during infrastructure analysis in Sprint 2
High false positive rates when correlating HIDS and NIDS data across multiple tools	Medium	High	Start with a limited scope for correlation rules and expand iteratively
Scope creep due to the combination of research, architecture design, and implementation	High	High	Prioritize the reference architecture deliverable and treat full implementation as best-effort
Limited access to RobotLab infrastructure making validation difficult	High	Medium	Set up a local lab environment early to simulate the RobotLab setup

11 Validation Strategy

The validation strategy for this research, would be divided in several steps:

- **Controlled Attack Simulation** - Run controlled attack scenarios, in order to validate the Advanced Threat Monitoring System is correctly put in place, as well as working efficiently. This should be done in an isolated lab environment, which simulates the real environment, to verify the Monitoring System is raising the right and expected alerts.
- **Detection Coverage Analysis** - Map the implemented monitoring against a threat framework. For each relevant attack technique applied, the outcome should

be documented based on whether the threat was detected fully, partially detected or if it was entirely missed.

- **False Positive/Negative Rate Measurement** - Run the standard expected workload through the system if and, if so, how many false alerts were generated, since the monitoring system won't be usable and reliable if it generates too much noise that needs to be filtered by its manager.
- **Comparison Against the Original Problem** - As of the end of the research and/or PoC implementation, we should compare with the initially risen problem, to verify which parts of it were addressed and which were not.

12 Conclusion

This research plan presents the approach for investigating advanced threat monitoring within the RobotLab environment. The current monitoring solution mainly relies on AWS GuardDuty, which provides limited visibility into low-level system behavior and network traffic.

The goal of this project is to research and design a monitoring architecture that integrates HIDS, NIDS, and SDN-based mechanisms to improve threat detection and response capabilities. By combining techniques such as eBPF-based monitoring, deep packet inspection, and centralized data correlation, the project aims to provide deeper visibility into the infrastructure.

Using the DOT Framework, the research will combine literature study, infrastructure analysis, laboratory testing, and stakeholder validation. The expected outcome is a reference architecture and a prototype implementation that demonstrates a more comprehensive threat monitoring approach for RobotLab.